



HA NOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Computer Vision

Chapter 3.2: Image transform

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Computer Vision

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Content

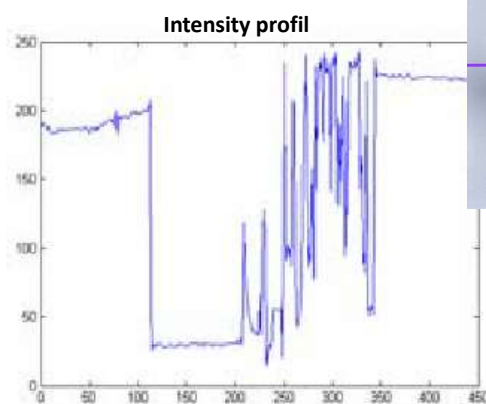
- Rappel: digital image representation
- Point Processing
- Convolution and Linear filtering
- More neighborhood operators
- Image transforms
 - Frequency domain
 - Frequencies in images
 - Fourier transform
 - Frequential Processing (frequential filters)
 - PCA (additional reading)



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Frequencies in images

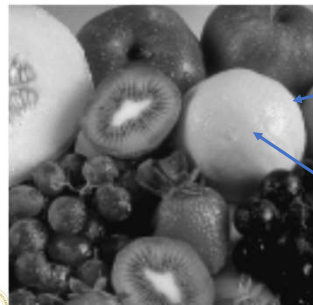


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Frequencies in images

- What are the (low/high) frequencies in an image?
 - Frequency = **intensity change**
 - Slow changes (homogeneous /blur regions): **low frequency**
 - fast/abrupt changes (egde, contour, noise): **high frequency**



High frequency

Low frequency

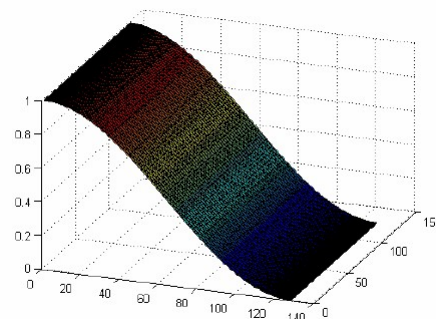
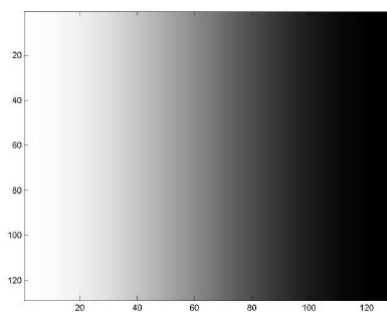
Most of energy concentrated in low frequencies



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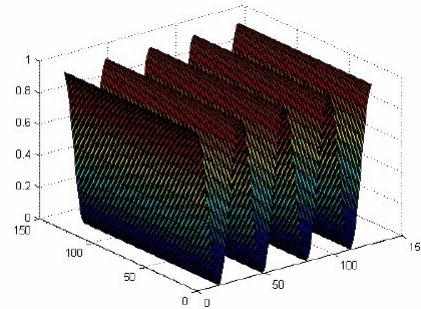
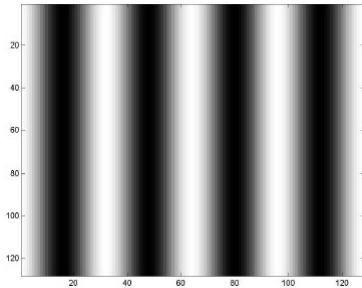
Low frequencies



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High frequencies



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Image spectral analysis

- An image is a visual signal
 - We can analyse the frequencies of the signal
- How?
 - we will create a new « image » which contains all frequencies of the image
 - Like a 2D frequency graphic
 - The basic tool for it is the **Fourier Transform**
- We talk about the **frequency domain**, opposing to the **spatial domain** (image)



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Frequencies in a signal

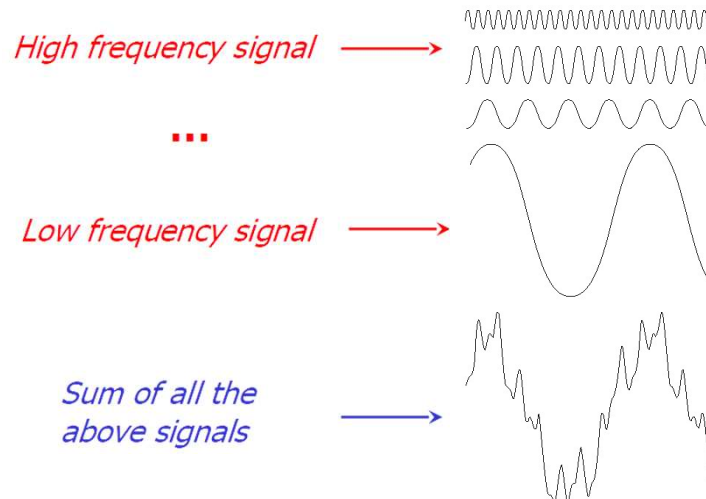


FIGURE 4.1 The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.



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Source : Gonzalez and Woods, *Digital Image Processing*, Prentice-Hall, 2002.

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Fourier series

A bold idea (1807) - Jean

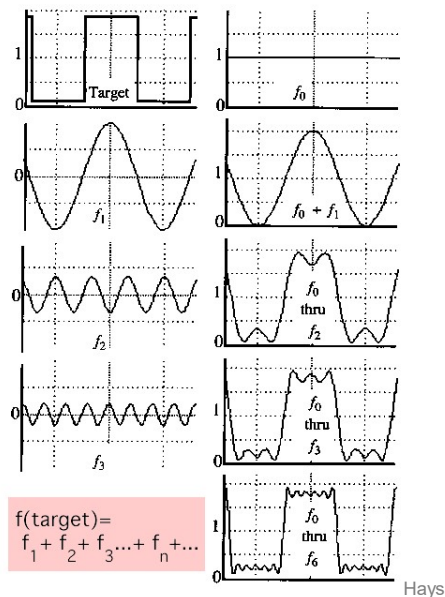
Baptiste Joseph Fourier (1768-1830):

Any univariate function can be rewritten as a **weighted sum of sines and cosines of different frequencies**.

Our building block:

$$A \sin(\omega t) + B \cos(\omega t)$$

Add enough of them to get any signal $g(t)$ you want!



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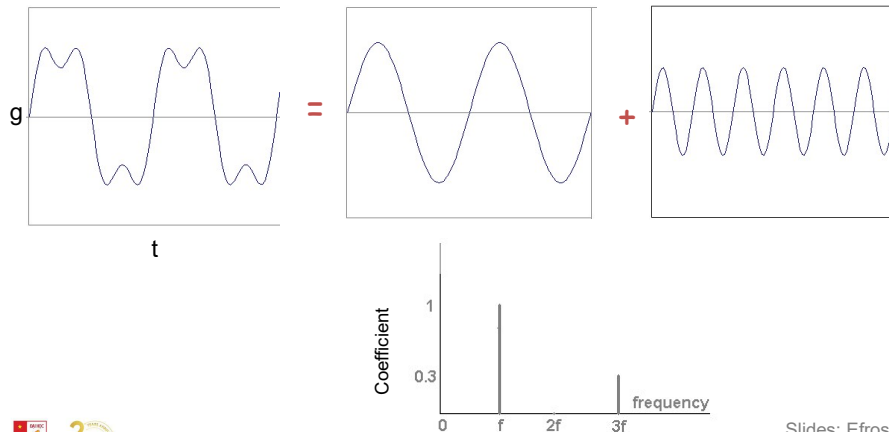
Hays

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Example

$$t = [0, 2], f = 1$$

$$g(t) = \sin(2\pi f t) + (1/3)\sin(2\pi(3f) t)$$



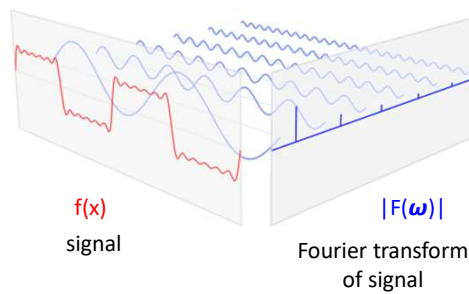
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Slides: Efros

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Fourier Transform

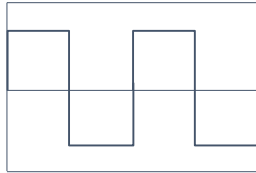
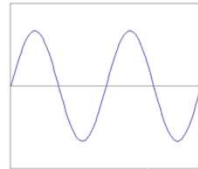
- Fourier transform is a mathematical transform that
 - Decomposes functions depending on space or time into functions depending on spatial or temporal frequency



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Fourier Series


 $\approx$


We want to get this function

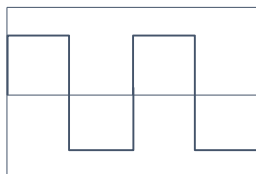
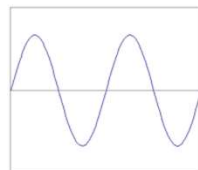
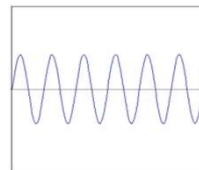


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Slide by Alexei A. Efros

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Fourier Series


 $\approx$

 $+$

 $=$


We want to get this function

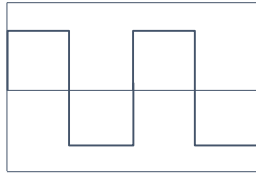


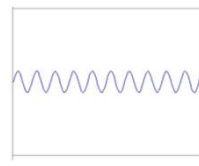
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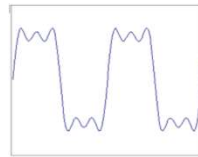
14

Fourier Series


 $\approx$

 $+$


We want to get this function

 $=$


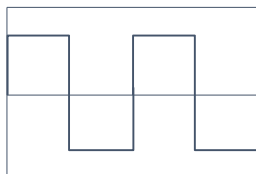
Slide by Alexei A. Efros

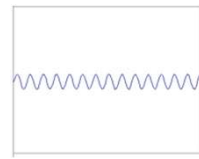


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Fourier Series


 $\approx$

 $+$


We want to get this function

 $=$

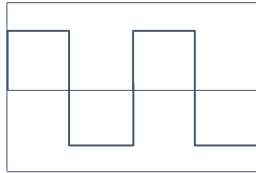

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Fourier Series


 $\approx$

 $+$


We want to get this function

 $=$

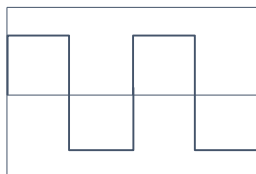

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Fourier Series


 $\approx$

 $+$


We want to get this function

 $=$

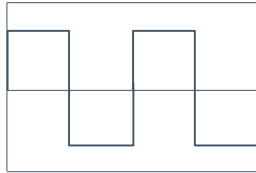

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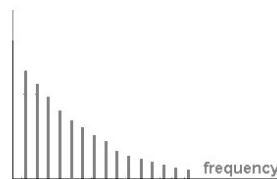
Fourier Series



We want to get this function

$$= A \sum_{k=1}^{\infty} \frac{1}{k} \sin(2\pi kt)$$

We'll get there in the limit



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The math

$$\text{Fourier Transform : } F(\omega) = \int_{-\infty}^{+\infty} f(x) e^{-i\omega x} dx$$

$$\text{Inverse Fourier Transform : } f(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{i\omega x} d\omega$$

- Where are the sines and cosines? $e^{i\omega x} = \cos(\omega x) + i \sin(\omega x)$
- The result is a complex function $F(\omega) = R(\omega) + iI(\omega)$
- We've been showing only the **amplitude A (spectre)** so far:
- Phase is also encoded:

$$A = \pm \sqrt{R(\omega)^2 + I(\omega)^2}$$

$$\phi = \tan^{-1} \frac{I(\omega)}{R(\omega)}$$



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Discrete Fourier transform

$$H_{f_j} = \frac{1}{N} \sum_k h_{t_k} e^{2\pi i f_j t_k}$$

$$h_{t_j} = \frac{1}{N} \sum_k H_{f_k} e^{-2\pi i f_k t_j}$$

where the t_k are the time corresponding to my signal in the time domain h_{t_k} , f_k are the corresponding frequency to my signal in the frequency domain, and N is the number of points of the signal data.

Fourier Transform Equation Explained:

https://www.youtube.com/watch?v=8V6Hi-kP9EE&ab_channel=IainExplainsSignals%2CSystems%2CandDigitalComms



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Magnitude and phase

- Fourier transform stores the **magnitude** and **phase** at each frequency
 - Magnitude **encodes how much signal there is** at a particular frequency
 - Phase **encodes spatial** information (indirectly)
 - For mathematical convenience, this is often notated in terms of real and complex numbers

Amplitude:

$$A = \pm \sqrt{R(\omega)^2 + I(\omega)^2}$$

Phase:

$$\phi = \tan^{-1} \frac{I(\omega)}{R(\omega)}$$

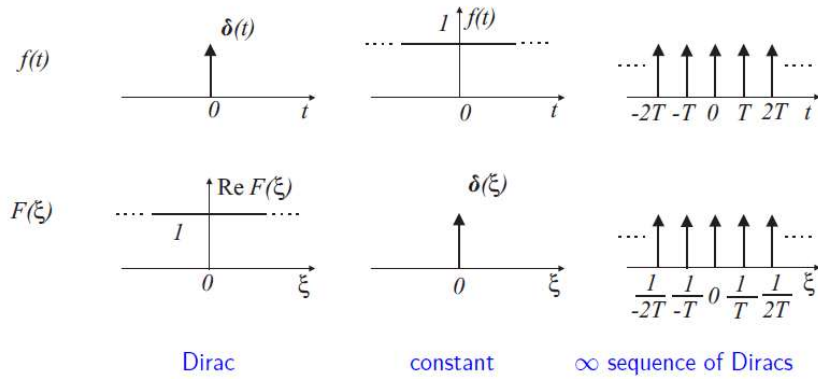
Slide by Rober Pless



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Basic Fourier Transform pairs

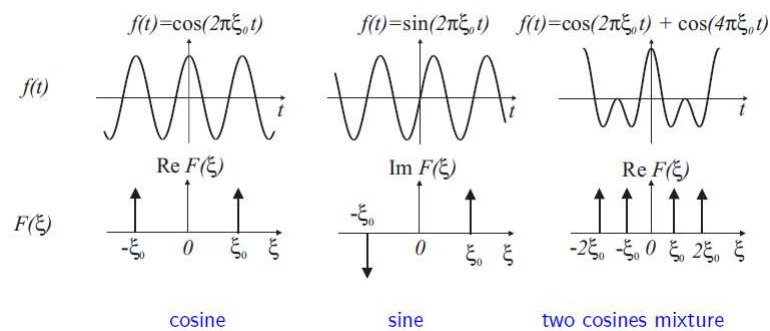


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Source: Václav Hlaváč - *Fourier transform, in 1D and in 2D*

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Basic Fourier Transform pairs

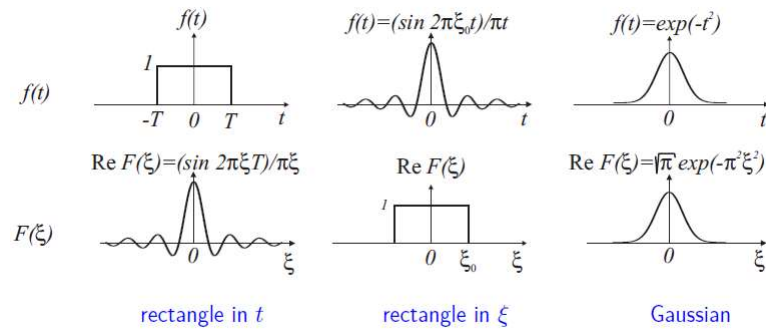


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Source: Václav Hlaváč - *Fourier transform, in 1D and in 2D*

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Basic Fourier Transform pairs



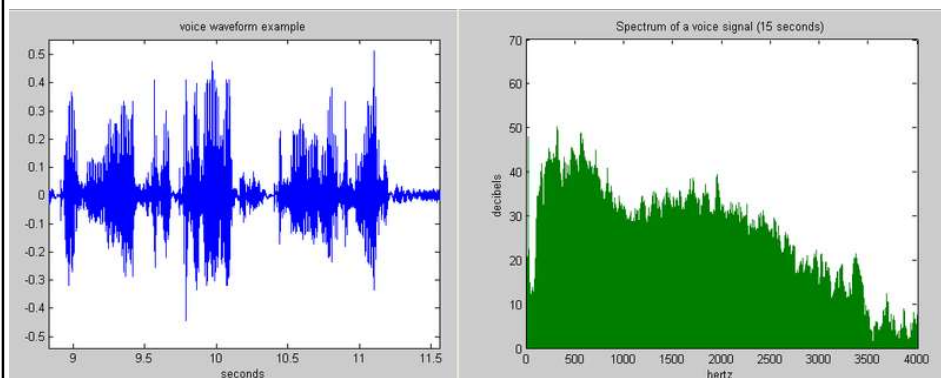
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Source: Václav Hlaváč - *Fourier transform, in 1D and in 2D*

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Example: Music

- We think of music in terms of frequencies at different magnitudes



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Slide: Hoiem

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2D FFT - discrete

Direct transform

$$F(u, v) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) \exp \left[-2\pi i \left(\frac{mu}{M} + \frac{nv}{N} \right) \right],$$

$$u = 0, 1, \dots, M-1, \quad v = 0, 1, \dots, N-1,$$

Inverse transform

$$f(m, n) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) \exp \left[2\pi i \left(\frac{mu}{M} + \frac{nv}{N} \right) \right],$$

$$m = 0, 1, \dots, M-1, \quad n = 0, 1, \dots, N-1.$$



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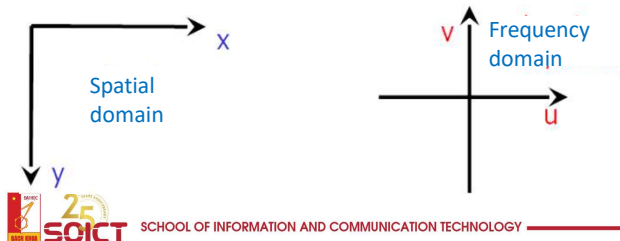
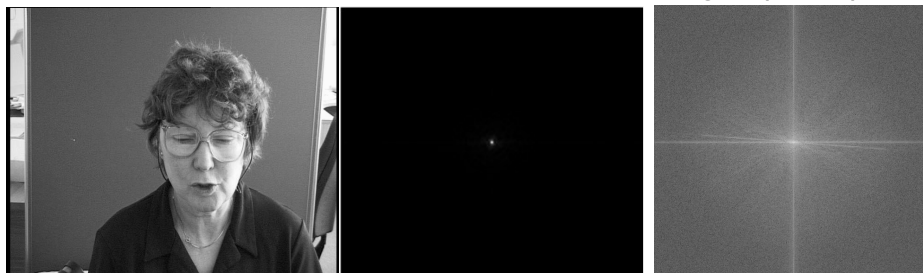
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Image Fourier transform

Original image

Spectra $|F(u, v)|$

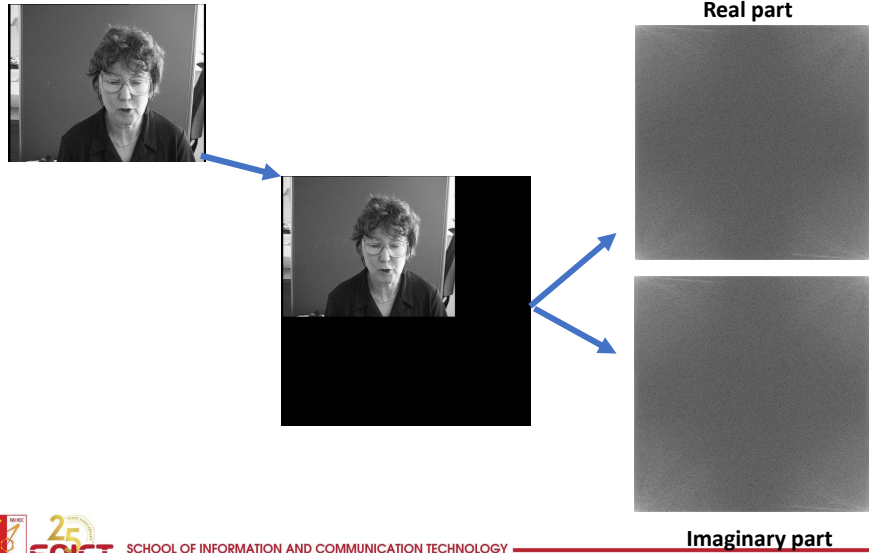
Enhanced Spectra
 $\log(1 + |F(u, v)|)$



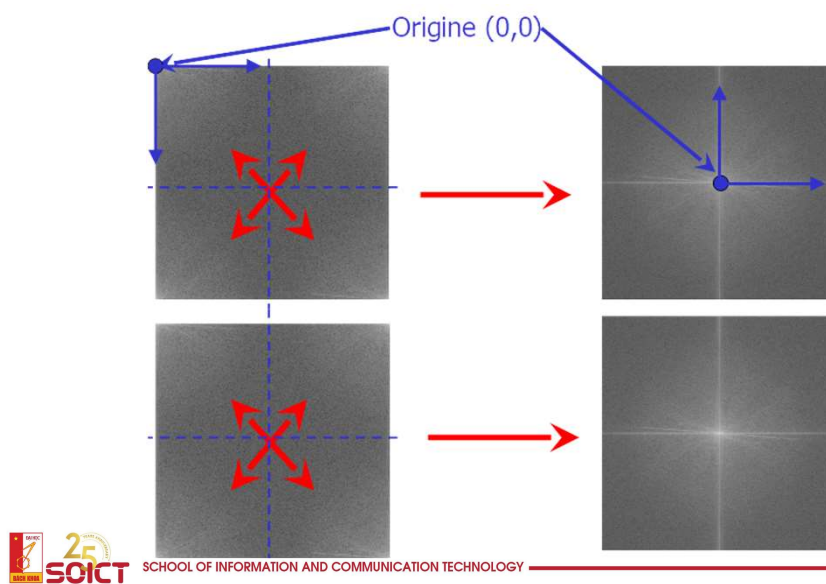
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FFT



FFT



Inverse FFT

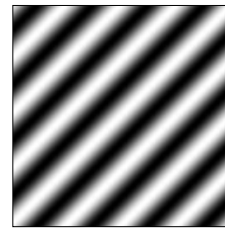
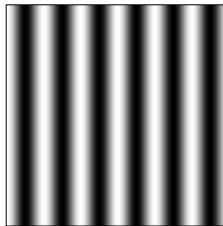
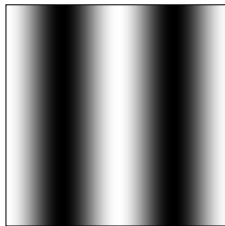


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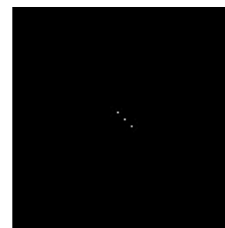
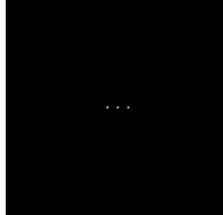
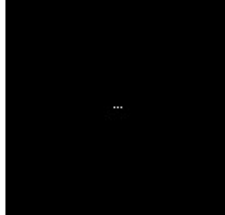
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Fourier analysis in images

Intensity images (spatial domain)



Fourier images (spectral images – amplitude images)



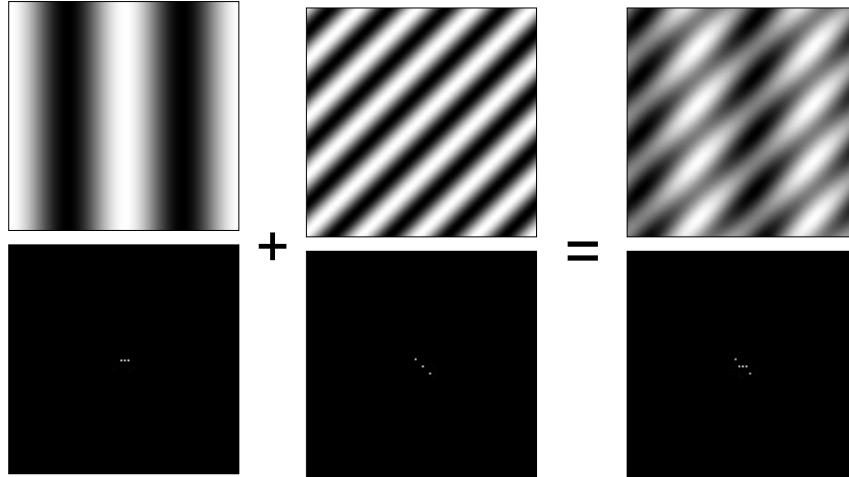
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<http://sharp.bu.edu/~slehar/fourier/fourier.html#filtering> More: <http://www.cs.unm.edu/~brayer/vision/fourier.html>

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Signals can be composed

Intensity images (spatial domain)



Fourier images (spectral images – amplitude images)

<http://sharp.bu.edu/~slehar/fourier/fourier.html#filtering>



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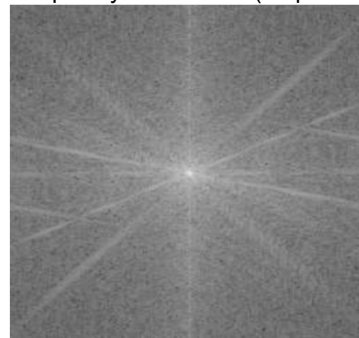
Fourier Transform of an image

Natural image



$f(x,y)$

Fourier decomposition
Frequency coefficients (amplitude)



$|F(\omega)|$

What does it mean to be at pixel x,y ?

What does it mean to be more or less bright in the Fourier decomposition image?



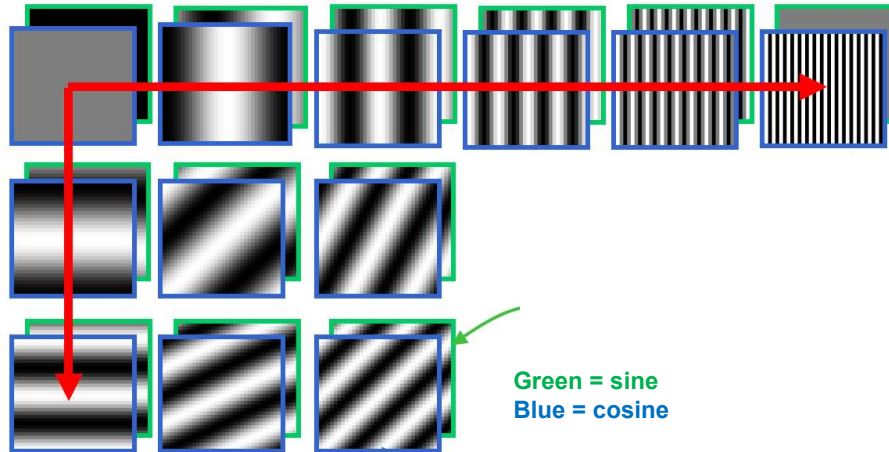
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Fourier Bases

Teases away 'fast vs. slow' changes in the image.

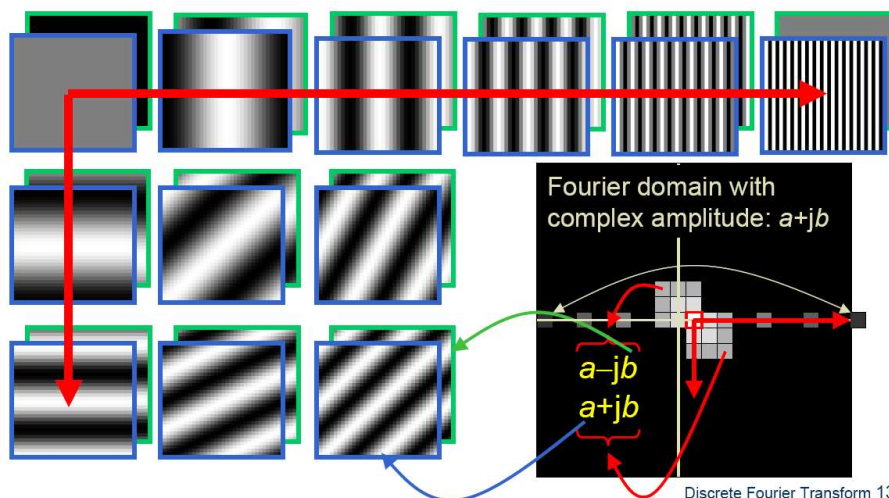


This change of basis is the Fourier Transform

Hays

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Fourier Bases

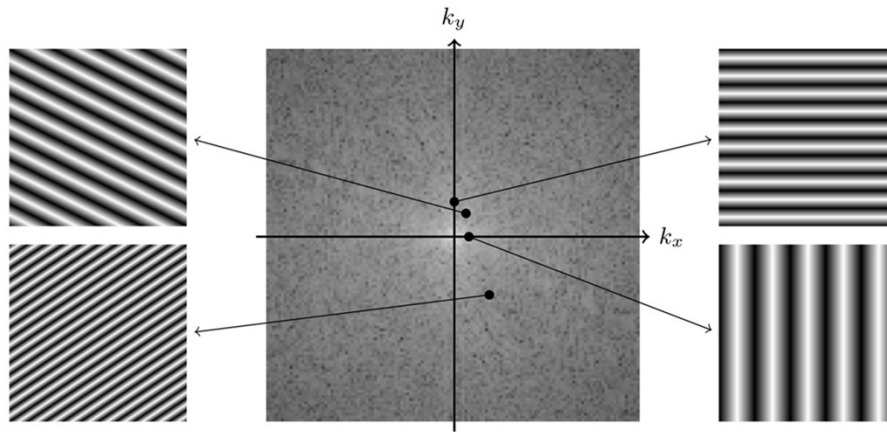


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Hays

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2D Fourier Transform



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Basis reconstruction



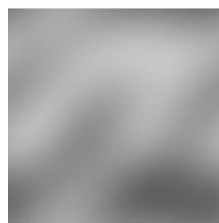
Full image



First 1 basis fn



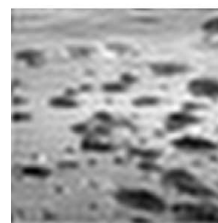
First 4 basis fns



First 9 basis fns



First 16 basis fns



First 400 basis fns



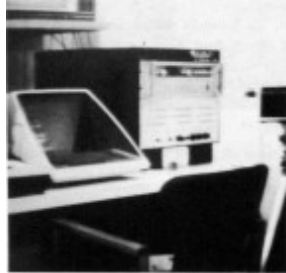
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Danny Alexander

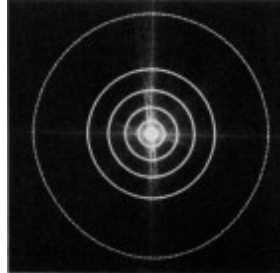
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2D Fourier transform

Image



Fourier Spectrum



Percentage of image power enclosed in circles (small to large) :
90, 95, 98, 99, 99.5, 99.9

Most of energy concentrated in low frequencies

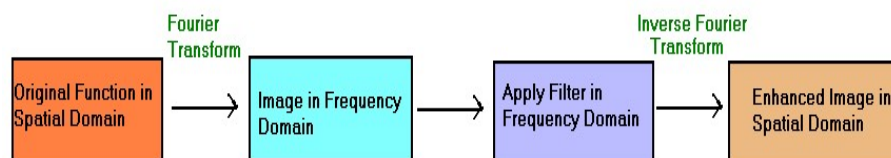


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Image filtering in the frequential domain

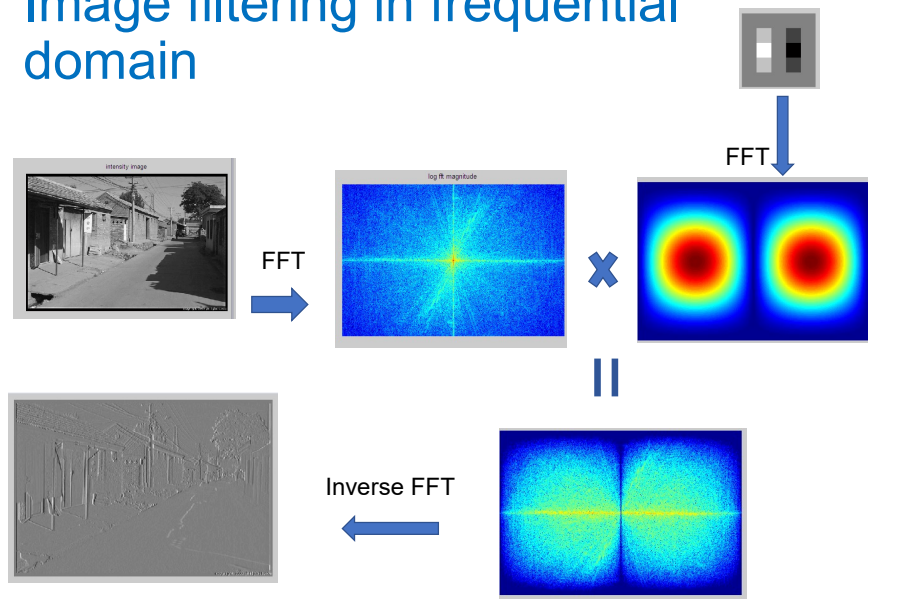
- We will be dealing only with functions (images) of finite duration so we will be interested only in Fourier Transform



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Image filtering in frequential domain

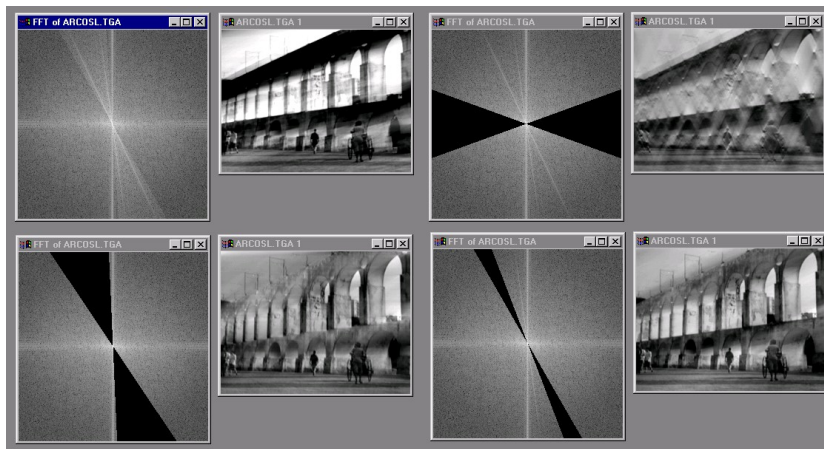


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Slide by Derek Hoiem

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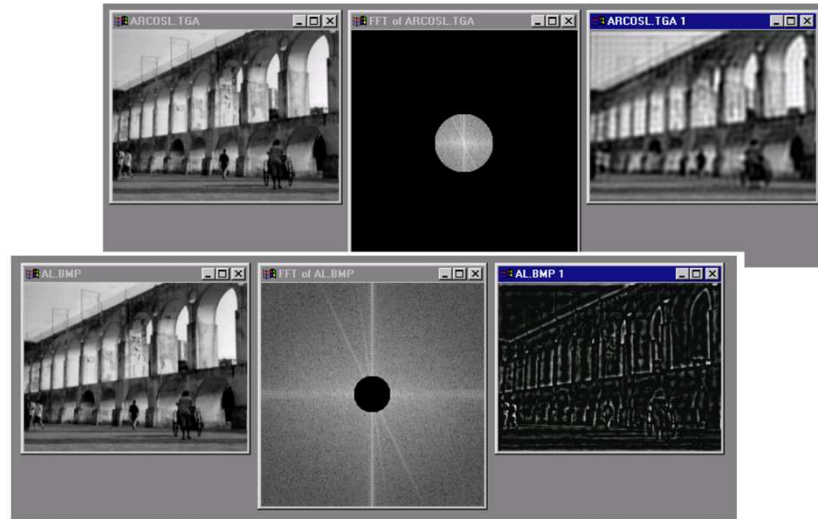
Now we can edit frequencies!



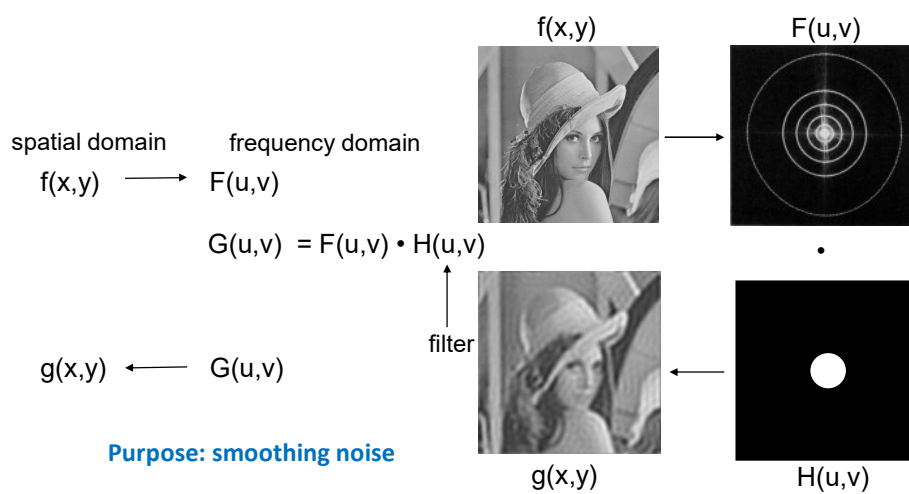
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Low-pass and high-pass filtering



Low-pass filter

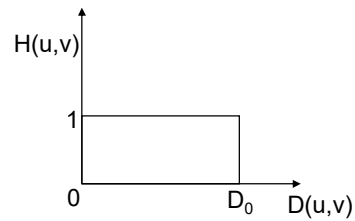
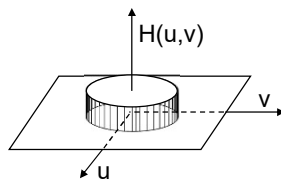


$H(u,v)$ - Ideal low-pass filter

$$H(u,v) = \begin{cases} 1 & D(u,v) \leq D_0 \\ 0 & D(u,v) > D_0 \end{cases}$$

$$D(u,v) = \sqrt{u^2 + v^2}$$

$$D_0 = \text{cut off frequency}$$



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Blurring - Ideal low-pass filters



99.7%

99.37%

98.65%



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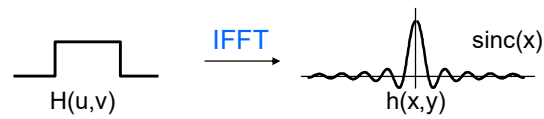
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The ringing problem

$$G(u,v) = F(u,v) \cdot H(u,v)$$

Convolution Theorem

$$g(x,y) = f(x,y) * h(x,y)$$



$\uparrow D_0 \longrightarrow \downarrow \text{Ringing radius} + \downarrow \text{blur}$

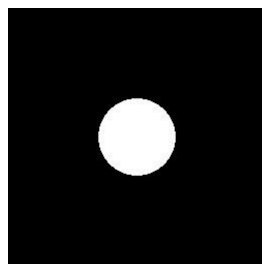


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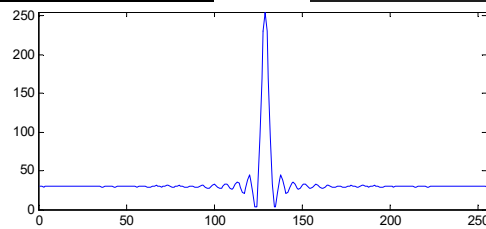
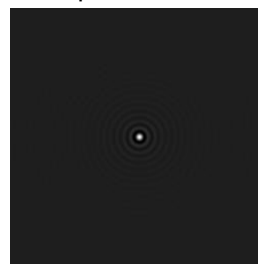
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The ringing problem

Freq. domain



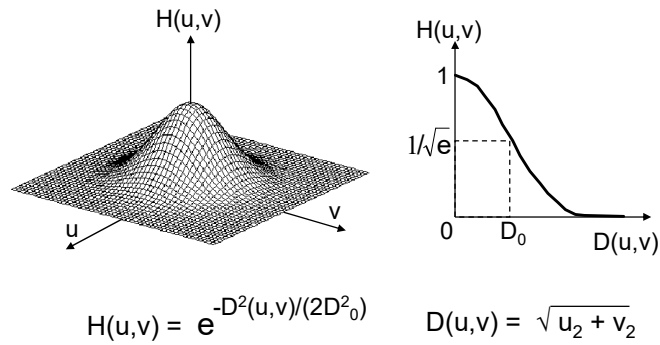
Spatial domain



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$H(u,v)$ - Gaussian filter



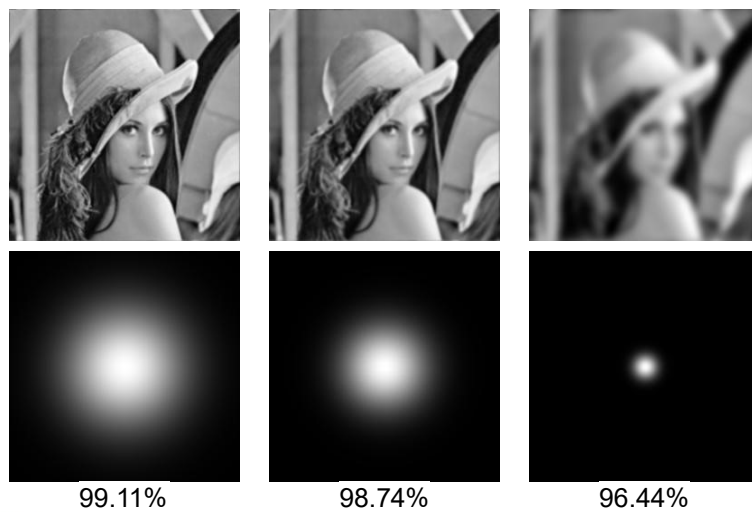
Softer Blurring + no Ringing



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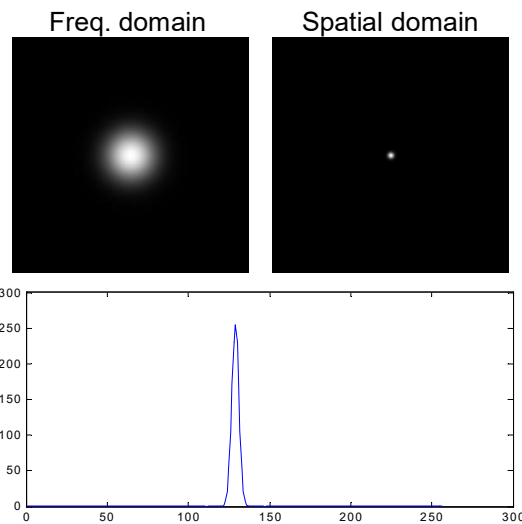
Blurring - Gaussain lowpass filter



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The Gaussian lowpass filter

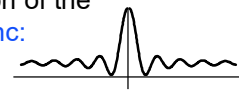


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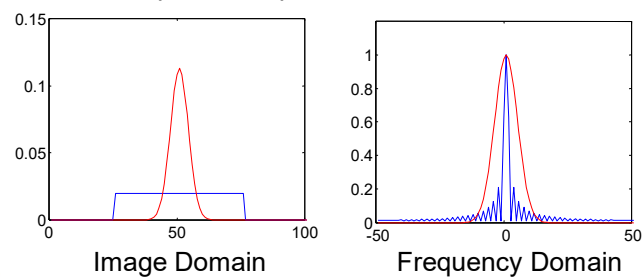
Blurring in the Spatial Domain

Averaging = convolution with $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ = point multiplication of the transform with **sinc**:



Gaussian Averaging = convolution with $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

= point multiplication of the transform with **a gaussian**.



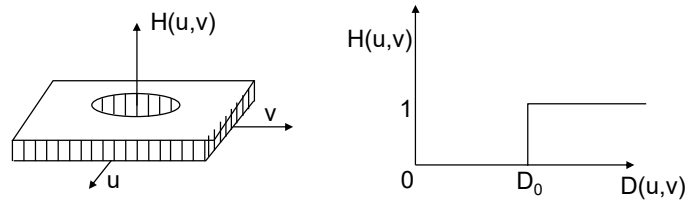
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High-pass filter

$H(u,v)$ - Ideal Filter

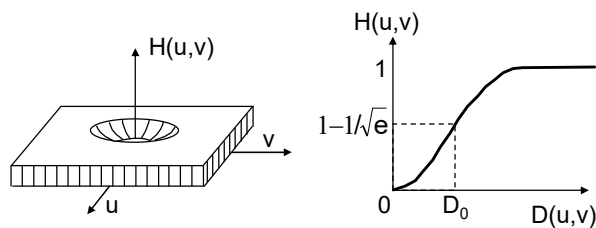
$$H(u,v) = \begin{cases} 0 & D(u,v) \leq D_0 \\ 1 & D(u,v) > D_0 \end{cases} \quad \begin{array}{l} D(u,v) = \sqrt{u^2 + v^2} \\ D_0 = \text{cut off frequency} \end{array}$$



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High-pass gaussian filter



$$H(u,v) = 1 - e^{-D^2(u,v)/(2D_0^2)}$$

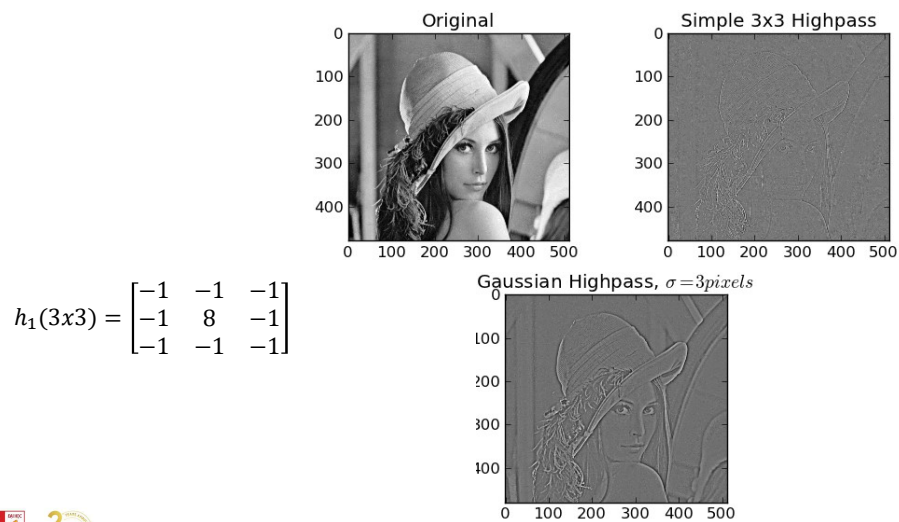
$$D(u,v) = \sqrt{u^2 + v^2}$$



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High-pass filtering



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High pass filtering

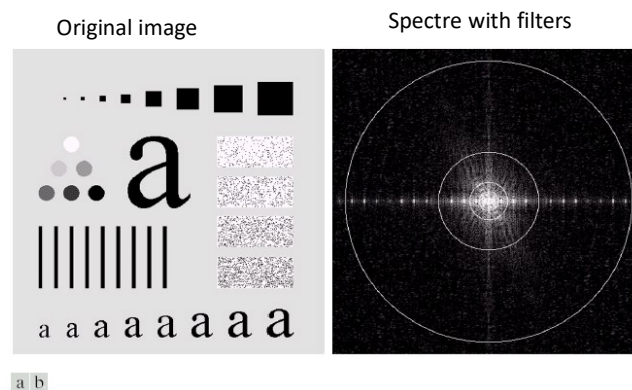


FIGURE 4.11 (a) An image of size 500×500 pixels and (b) its Fourier spectrum. The superimposed circles have radii values of 5, 15, 30, 80, and 230, which enclose 92.0, 94.6, 96.4, 98.0, and 99.5% of the image power, respectively.



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Source : Gonzalez and Woods. *Digital Image Processing*. Prentice-Hall, 2002.

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High pass filtering

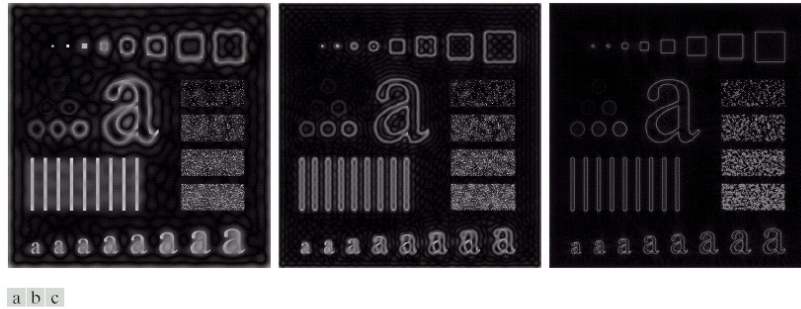


FIGURE 4.24 Results of ideal highpass filtering the image in Fig. 4.11(a) with $D_0 = 15, 30$, and 80 , respectively. Problems with ringing are quite evident in (a) and (b).



Source : Gonzalez and Woods. *Digital Image Processing*. Prentice-Hall, 2002.

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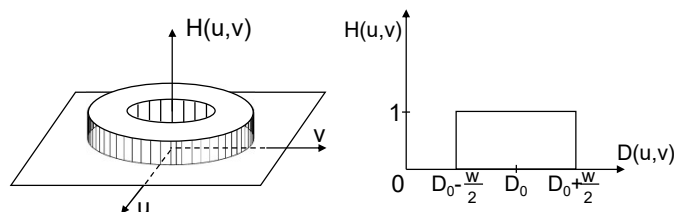
Band-pass filtering

$$H(u,v) = \begin{cases} 0 & D(u,v) \leq D_0 - \frac{w}{2} \\ 1 & D_0 - \frac{w}{2} \leq D(u,v) \leq D_0 + \frac{w}{2} \\ 0 & D(u,v) > D_0 + \frac{w}{2} \end{cases}$$

$$D(u,v) = \sqrt{u^2 + v^2}$$

D_0 = cut off frequency

w = band-width



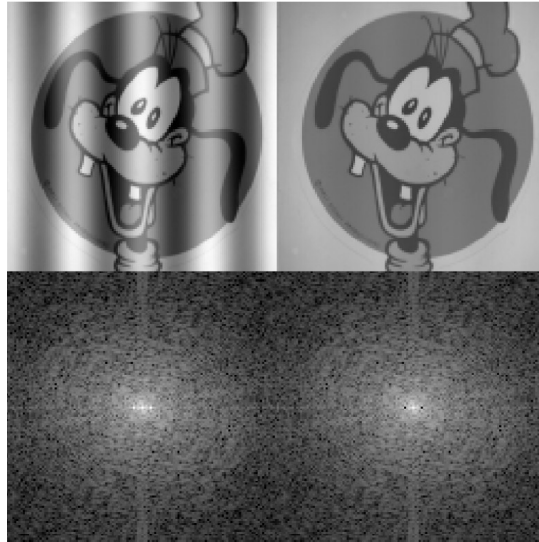
Can be obtained by multiplying the filter functions of a **low-pass** and of a **high-pass** in the frequency domain



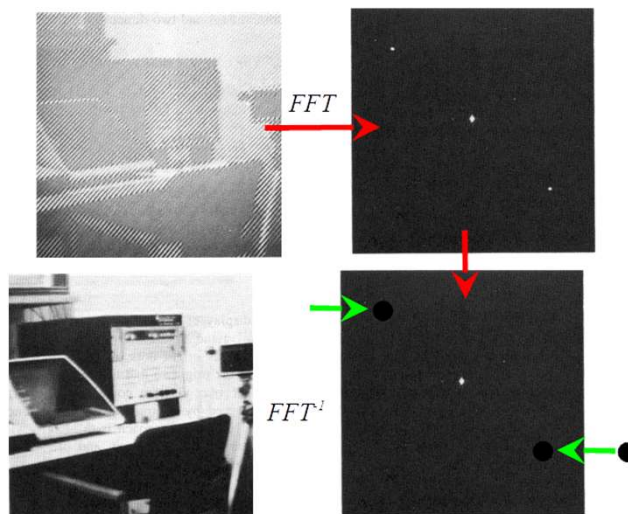
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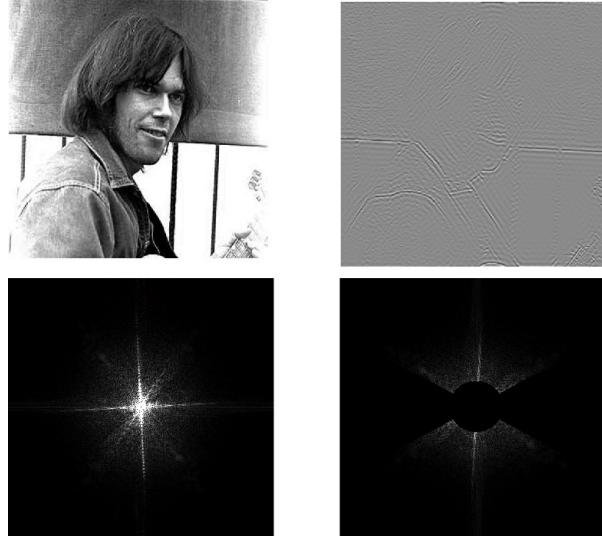
Removing sinus noise



Removing sinus noise



High-pass filtering + orientation



Hybrid Images

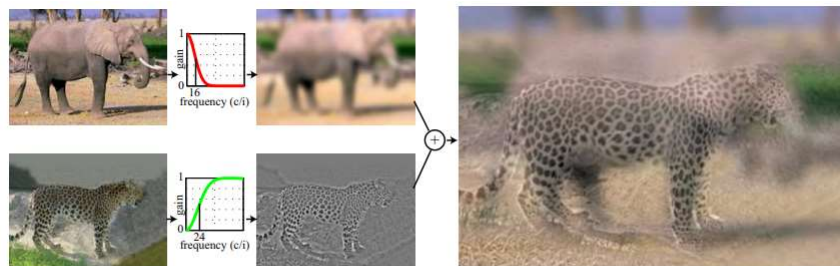


Figure 2: hybrid images are generated by superimposing two images at two different spatial scales: the low-spatial scale is obtained by filtering one image with a low-pass filter, and the high spatial scale is obtained by filtering a second image with a high-pass filter. The final hybrid image is composed by adding these two filtered images.

A. Oliva, A. Torralba, P.G. Schyns, SIGGRAPH 2006

Marylin Einstein



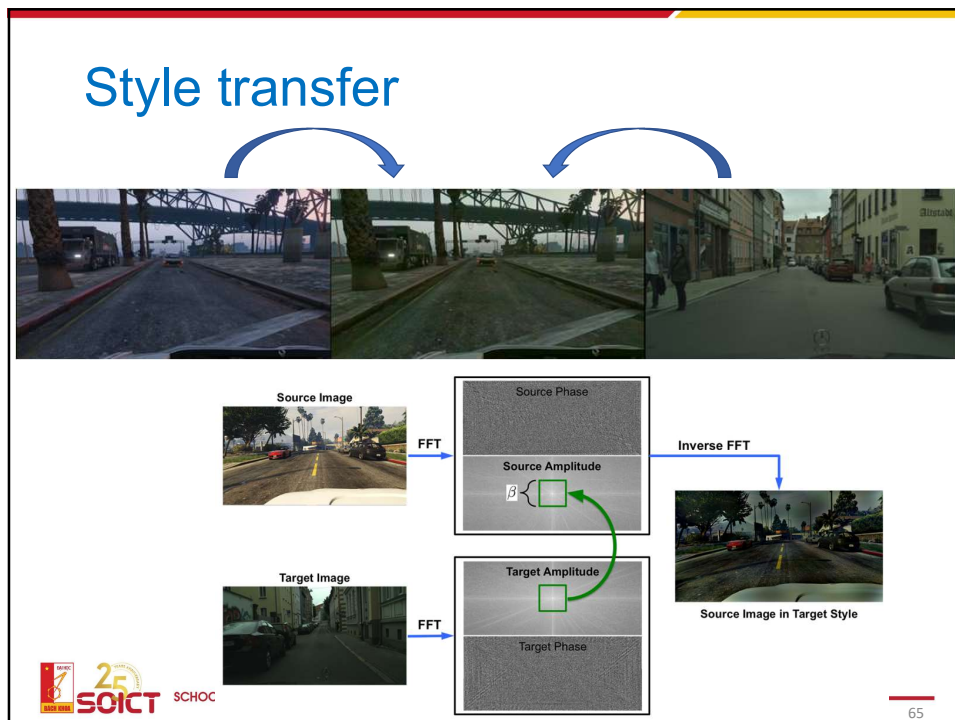
Copyright © 2007 Aude Oliva, MIT

When you look at the image above, whose face do you see? At normal screen viewing distance you should see the face of the great scientist Albert Einstein. Now squint your eyes or take a few steps back from the image. Does a certain Hollywood personality pop into view? The Marilyn Einstein hybrid image was created by Dr. Aude Oliva for the March 31st 2007 issue of New Scientist magazine.

http://cvcl.mit.edu/hybrid_gallery/monroe_einstein.html

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- Rappel: digital image representation
- Point Processing
- Convolution and Linear filtering
- More neighborhood operators
- Image transforms
 - Frequency domain
 - PCA (additional reading)
 - PCA
 - Example of using PCA for face recognition



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Principle Component Analysis - PCA (Karhunen-Loeve transformation)

- **PCA** transforms the original input space into a lower dimensional space
 - By constructing dimensions that are linear combinations of the given features
- The objective: consider **independent dimensions** along which data have **largest variance** (i.e., greatest variability)



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Principal Component Analysis (cont.)

- **PCA** enables transform a number of possibly correlated variables into a smaller number of uncorrelated variables called **principal components**
- The **first principal component** accounts for as much of the **variability** in the data as possible
- Each **succeeding component** (orthogonal to the previous ones) accounts for as much of the remaining variability as possible



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Principal Component Analysis (cont.)

- PCA is the most commonly used dimension reduction technique.

- Data samples

$$x_1, \dots, x_N$$

- Compute the mean

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

- Computer the covariance matrix:

$$\Sigma_x = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})(x_i - \bar{x})^T$$



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Principal Component Analysis (cont.)

- Compute the eigenvalues λ and eigenvectors e of the matrix Σ_x
- Solve $\Sigma_x x = \lambda x$
- Order them by magnitude:

$$\lambda_1 \geq \lambda_2 \geq \dots \lambda_N.$$
- PCA reduces the dimension by keeping direction e such that $\lambda > T$.



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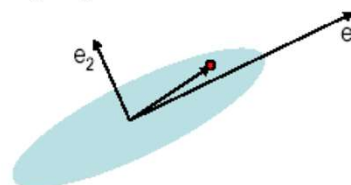
Principal Component Analysis (cont.)

- For many datasets, most of the eigenvalues are negligible and can be discarded.

The eigenvalue λ measures the variation in the direction of corresponding eigenvector

Example:

$$\lambda_1 \neq 0, \lambda_2 = 0.$$



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Principal Component Analysis (cont.)

- How to get uncorrelated components which Capture most of the variance
- Project the data onto the selected eigenvectors:

$$y_i = e_i^T (x_i - \bar{x})$$
- If we consider first M eigenvectors we get new lower dimensional representation

$$[y_1, \dots, y_M]$$
- Proportion covered by first M eigenvalues

$$\frac{\sum_{i=1}^M \lambda_i}{\sum_{i=1}^N \lambda_i}$$

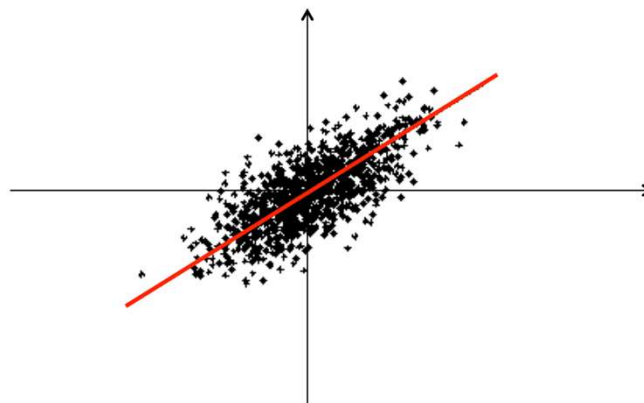
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Illustration of PCA



First principal component of a two-dimensional data set.

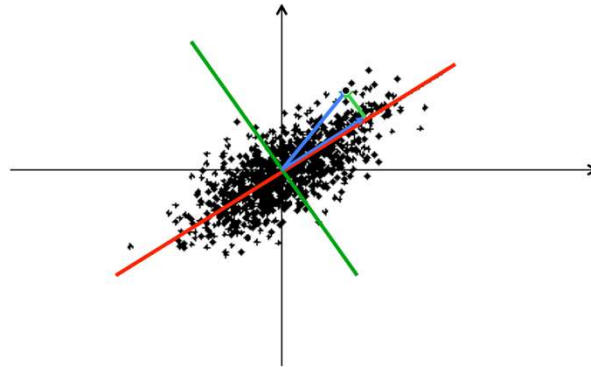
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Illustration of PCA



Second principal component of a two-dimensional data set.



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Determining the number of components

- Plot the eigenvalues
 - each eigenvalue is related to the amount of variation explained by the corresponding axis (eigenvector)
 - If the points on the graph tend to level out (show an “elbow” shape), these eigenvalues are **usually close enough to zero that they can be ignored**



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The space of all face images

- When viewed as **vectors of pixel values**, face images are extremely high-dimensional
 - 100x100 image = 10,000 dimensions
- However, relatively **few 10,000-dimensional vectors correspond to valid face images**
- We want to effectively **model the subspace** of face images



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Eigenfaces: Key idea

- Assume that most face images lie on a **low-dimensional subspace determined by the first k ($k < d$)** directions of maximum variance
- Use PCA to determine the vectors or “**eigenfaces**” **$u_1, \dots, u_k$** that span that subspace
- Represent all face images in the dataset as **linear combinations of eigenfaces**

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Eigenfaces example- Training images



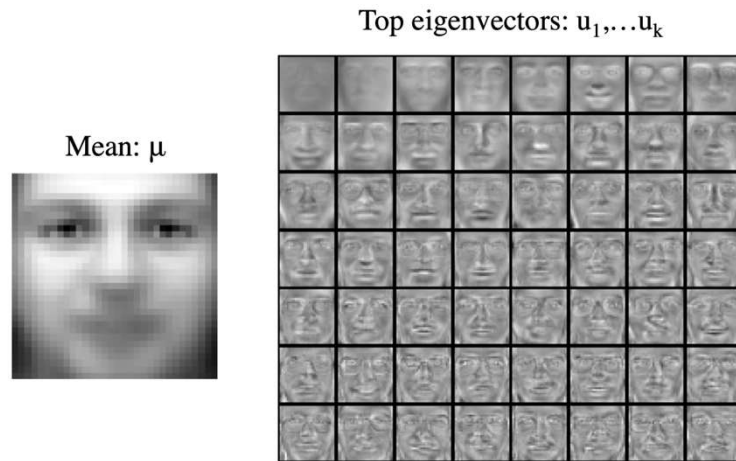
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Eigenfaces example



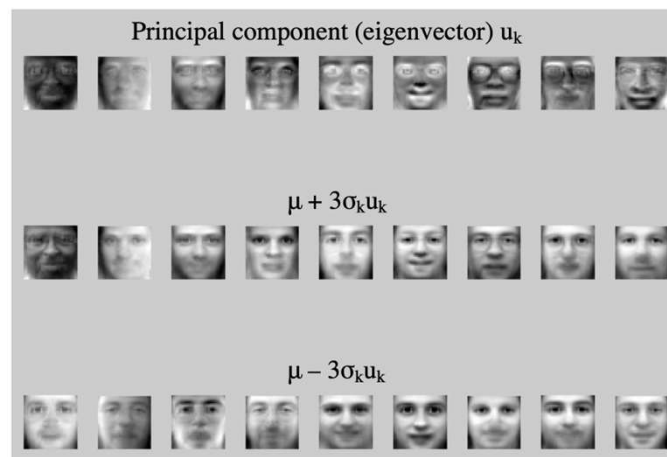
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Eigenfaces example



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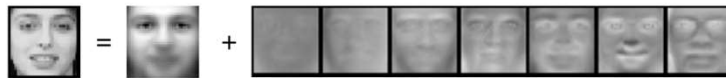
Eigenfaces examples

- Representation



$$(w_{i1}, \dots, w_{ik}) = (\mathbf{u}_1^T(\mathbf{x}_i - \boldsymbol{\mu}), \dots, \mathbf{u}_k^T(\mathbf{x}_i - \boldsymbol{\mu}))$$

- Reconstruction



$$\hat{\mathbf{x}} = \boldsymbol{\mu} + w_1 \mathbf{u}_1 + w_2 \mathbf{u}_2 + w_3 \mathbf{u}_3 + w_4 \mathbf{u}_4 + \dots$$



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Recognition with eigenfaces

- Process labeled training images:
- Find mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$
- Find k principal components (eigenvectors of $\boldsymbol{\Sigma}$) $\mathbf{u}_1, \dots, \mathbf{u}_k$
- Project each training image $\mathbf{x}_i$ onto subspace spanned by principal components:
 $(w_{i1}, \dots, w_{ik}) = (\mathbf{u}_1^T(\mathbf{x}_i - \boldsymbol{\mu}), \dots, \mathbf{u}_k^T(\mathbf{x}_i - \boldsymbol{\mu}))$
- Given novel image $\mathbf{x}$:
- Project onto subspace:
 $(w_1, \dots, w_k) = (\mathbf{u}_1^T(\mathbf{x} - \boldsymbol{\mu}), \dots, \mathbf{u}_k^T(\mathbf{x} - \boldsymbol{\mu}))$
- Optional: check reconstruction error $\mathbf{x} - \hat{\mathbf{x}}$ to determine whether image is really a face
- Classify as closest training face in k -dimensional subspace



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